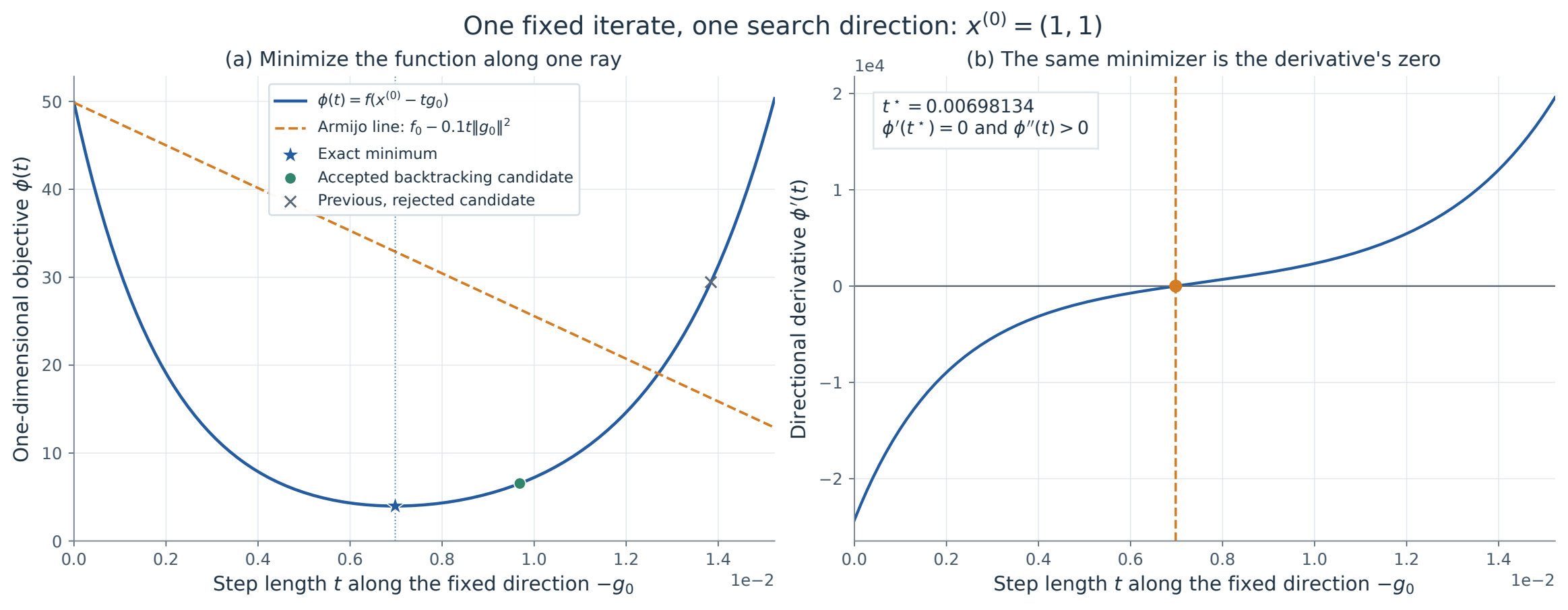
One fixed iterate, one search direction: $x^{(0)} = (1, 1)$
(a) Minimize the function along one ray
$\phi(t) = f(x^{(0)} - tg_0)$
Armijo line: $f_0 - 0.1t\|g_0\|^2$
Exact minimum
Accepted backtracking candidate
Previous, rejected candidate
One-dimensional objective $\phi(t)$
Step length $t$ along the fixed direction $-g_0$
1e−2
(b) The same minimizer is the derivative's zero
$t^\star = 0.00698134$
$\phi'(t^\star) = 0$ and $\phi''(t) > 0$
1e4
Directional derivative $\phi'(t)$
Step length $t$ along the fixed direction $-g_0$
1e−2